

Maternal Health Risk Stratification – v3

Improved High-Risk Recall · Still 3-class · Pipeline + class_weight + GridSearch

Key result: High-risk Recall raised from ~0.61 to **0.78** while remaining a 3-class problem (Low / Mid / High). Accuracy 70%. Educational only – not a diagnostic tool.

What changed vs previous version

- **class_weight='balanced'** on both Logistic Regression and Decision Tree
- Limited **GridSearchCV** (max_depth 3–6, min_samples_leaf 3/5/8, gini/entropy) with scoring=**recall_macro**
- Best params: criterion=entropy, max_depth=4, min_samples_leaf=5
- Full sklearn Pipeline (StandardScaler inside LR) – no leakage
- Did **not** convert to binary High vs Rest (would inflate metrics but violate the brief)

Q1 – Population & cleaning

Raw 1014 rows → after removing 562 exact duplicates and 2 HeartRate=7 records → **451 unique**. RiskLevel as ordered category. Low 51.7%, High 24.8%, Mid 23.5%.

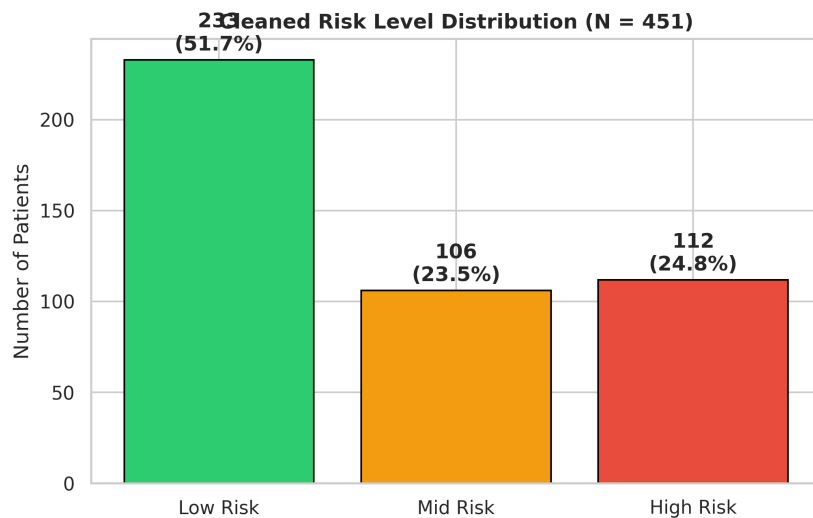


Figure 1. Cleaned risk distribution.

Q2 – Feature differences

BS and SystolicBP remain the strongest separators. High-risk median BS $\approx$ 11 mmol/L vs $\approx$ 7 in other groups.

Clinical Feature Distributions by Risk Level

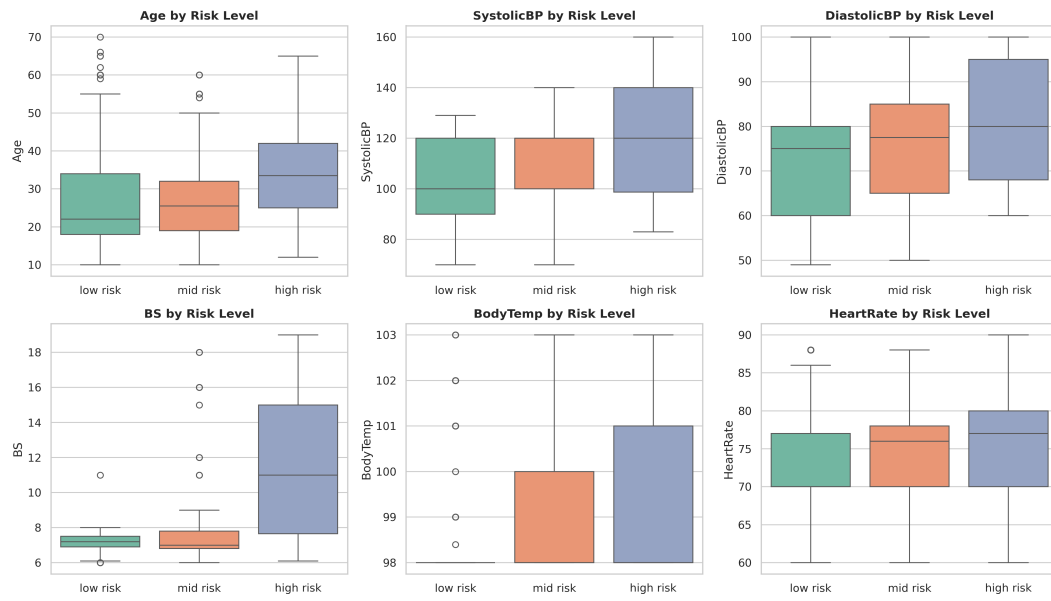


Figure 2. Box-plots by risk level.

Q3 – Association with High Risk

Pearson with High-Risk indicator: BS 0.57, SystolicBP 0.29, DiastolicBP 0.26, BodyTemp 0.22, Age 0.19, HeartRate 0.18.
Association $\neq$ causation.

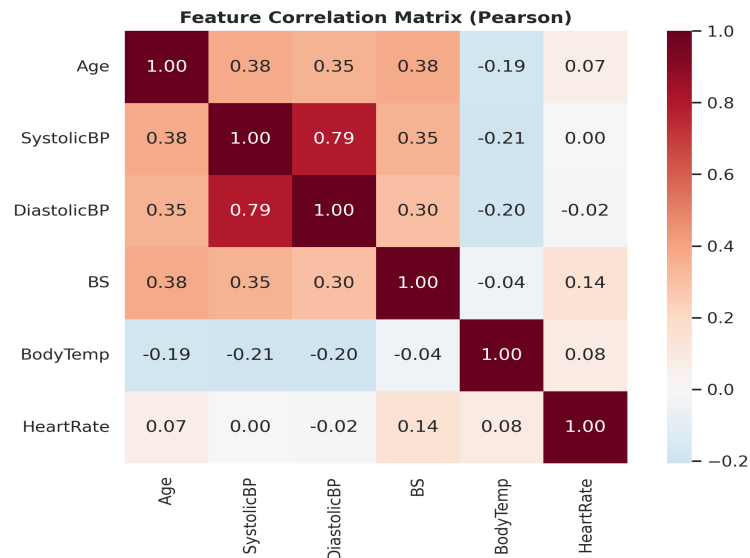


Figure 3. Correlation heatmap.

Q4 & Q5 – Models & metrics

Decision Tree (tuned, balanced): Accuracy 70.3%, High-risk Recall **0.78**, High-risk F1 0.77.

Logistic Regression (balanced): Accuracy 61.5%, High-risk Recall 0.57.

High-risk Recall improved substantially vs the previous untuned tree (~ 0.61). Mid-risk remains difficult (Recall still low) due to feature overlap.

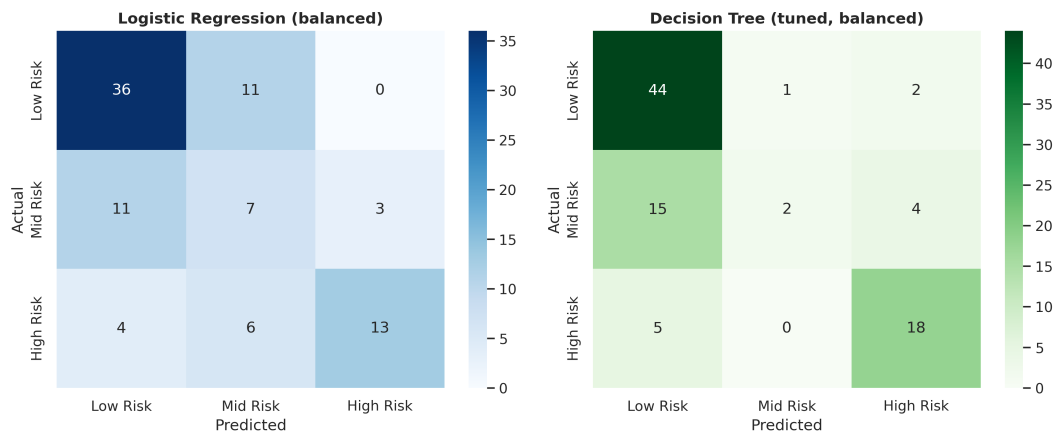


Figure 4. Confusion matrices.

Q6 – Errors

Dominant residual errors: Mid $\rightarrow$ Low and some High $\rightarrow$ Low. High $\rightarrow$ Low count dropped (better Recall) compared with the untuned model.

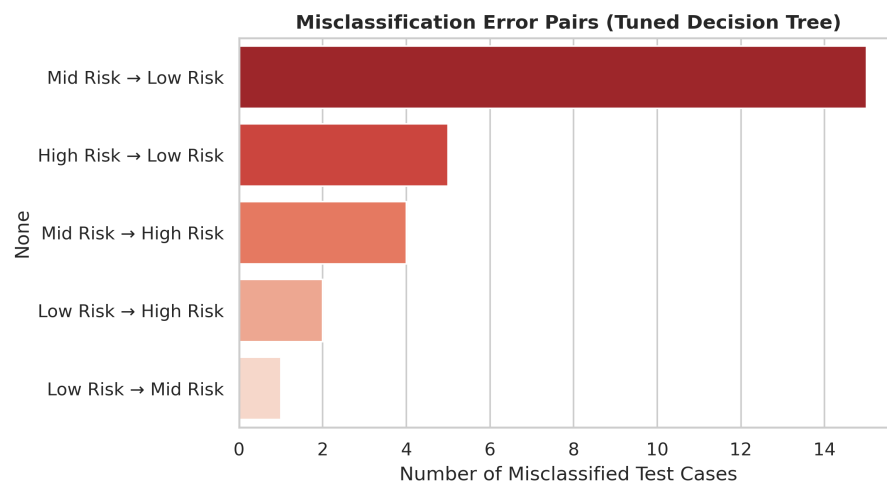


Figure 5. Error pairs.

Q7 – Drivers

DT Gini: BS ~47%, SystolicBP ~30%, BodyTemp ~12%, Age ~8%. LR coefficients for High-risk also rank BS highest (positive).

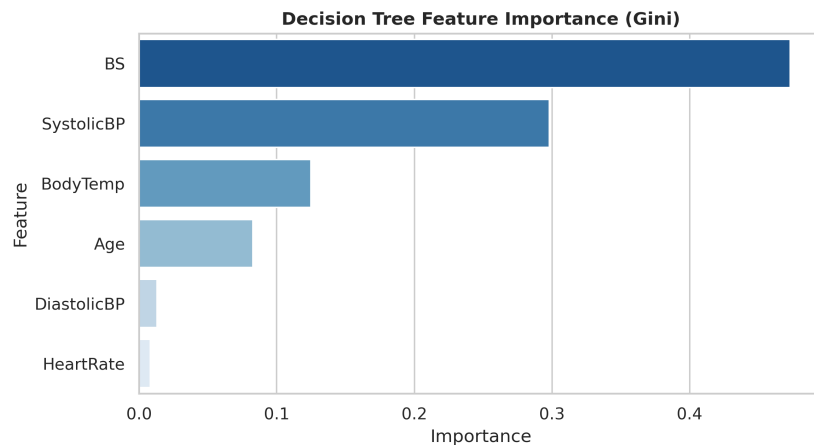


Figure 6. DT feature importance.

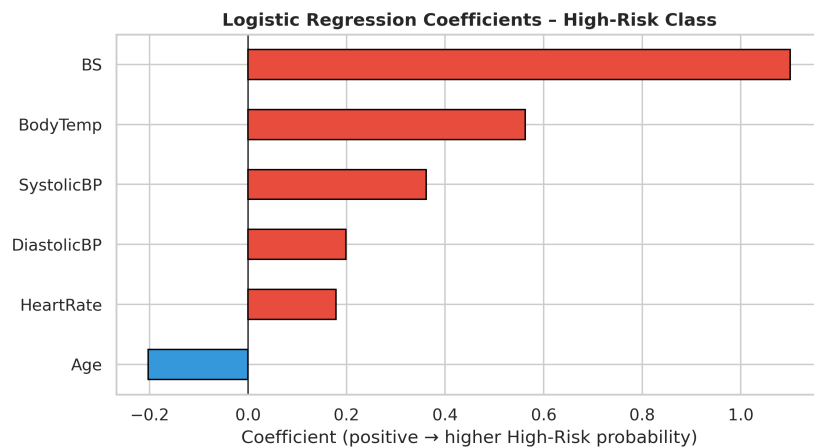


Figure 7. LR coefficients (High-risk class).

Q8 – Take-aways

1. BS and SystolicBP are the primary biomarkers in this dataset.
2. With balanced weights + light tuning we reach High-risk Recall 0.78 while staying 3-class.
3. Model is still only a preliminary screening aid under clinician supervision.
4. Limitations: original 55% duplicates, n=451, single geography, missing covariates (proteinuria, gestational age, BMI).
5. We deliberately did not switch to binary classification, which would look better on paper but would not answer the project questions about Mid risk.